ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ROBOTICS FUNDAMENTALS

PAPER 2 · PRACTICE B · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Robotics Fundamentals paper 2 · practice b.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Accept other correct answers, sensible design choices, equivalent code and neat, recognisable diagrams.

SECTION A

QUESTION 1

1.1 1.1.1 B ✓ 1.1.2 C ✓ 1.1.3 B ✓ 1.1.4 A ✓ 1.1.5 D ✓ 1.1.6 A ✓ 1.1.7 B ✓ 1.1.8 A ✓ 1.1.9 C ✓ 1.1.10 D ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 True ✓ 1.2.3 True ✓ 1.2.4 False ✓ 1.2.5 False ✓ 1.2.6 False ✓ 1.2.7 False ✓ 1.2.8 True ✓ 1.2.9 True ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 C ✓ 1.3.2 G ✓ 1.3.3 I ✓ 1.3.4 B ✓ 1.3.5 E ✓ 1.3.6 J ✓ 1.3.7 F ✓ 1.3.8 H ✓ 1.3.9 A ✓ 1.3.10 D ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 E.g. "Design and build a small autonomous robot that follows a black line on a white mat from start to finish as fast as possible and stops at the finish bar." – states the problem ✓ and what must be made ✓ (2)

2.2 Any FOUR ✓✓✓✓: fits within the size limit (e.g. 20 cm × 20 cm); costs less than a set budget; runs on batteries for the whole competition; follows curves without leaving the line; stops at the finish bar; started by a button only; uses the given parts (Arduino, two IR sensors). (4)

2.3 Inputs ✓✓: left IR sensor, right IR sensor, start button. Process ✓: Arduino Uno. Outputs ✓✓: motor driver to left and right DC motors, LED. Power supply / arrows in the correct direction and labels ✓ (6)

2.4 When the robot drifts, one sensor moves over the line and detects black ✓, telling the controller which way to turn back onto the line ✓. (2)

2.5 4 × AA cells (or a rechargeable pack) ✓ – they supply more current for longer, so the motors do not slow down quickly (a small 9 V battery runs flat fast under motor load) ✓ (2)

[16]

QUESTION 3

3.1 3.1.1 drive straight forward ✓✓; 3.1.2 turn left (the line is under the left sensor, so the robot has drifted right) ✓✓; 3.1.3 turn right ✓✓; 3.1.4 stop (finish bar) ✓✓ (8)

3.2 Measuring the sensor readings on the real mat under the real lighting ✓ and setting the black/white limit from them, because surfaces and light differ, so fixed values may not work ✓. (2)

3.3 $\text{Threshold} = (200 + 800) \div 2 \checkmark = 500 \checkmark$ (2)

3.4 The robot is too fast / turns too slowly, so it overshoots the line before it can correct ✓✓. Fixes (any TWO) ✓✓: reduce the speed (PWM); turn more sharply (stop or reverse the inside wheel); move the sensors further forward or closer together; add a third (middle) sensor; check the program loop has no long delays. (4)

[16]

QUESTION 4

4.1 Start ✓; "button pressed?" decision with loop back while not pressed ✓; read both sensors ✓; decision both black → stop motors and LED on, then end ✓; decision left black → turn left ✓; decision right black → turn right ✓; otherwise forward ✓; loop back to read the sensors / correct symbols ✓ (8)

4.2 Read both pins ✓✓; both HIGH → stopRobot() ✓; left HIGH → turnLeft() ✓; right HIGH → turnRight() ✓; else forward() ✓; correct if-else structure ✓ and syntax ✓, e.g. (8)

```
void loop() {
  int left = digitalRead(2);
  int right = digitalRead(3);
  if (left == HIGH && right == HIGH) {
    stopRobot();
  } else if (left == HIGH) {
    turnLeft();
  } else if (right == HIGH) {
    turnRight();
  } else {
    forward();
  }
}
```

4.3 A "when green flag / start" hat block ✓; a "forever" loop ✓; "if … then … else" blocks ✓ that test the sensor blocks (e.g. "left sensor = black") and contain motor blocks (forward, turn left, turn right, stop) ✓. (4)

4.4 A named block of code that does one job and can be called whenever it is needed ✓; it avoids repeating code and makes the program shorter, easier to read and easier to fix ✓. (2)

[22]

QUESTION 5

5.1 In order ✓✓✓✓: design / measure the chassis in CAD; export as an STL file; slice it (layer height, infill, supports) to make G-code; print it on the 3D printer; remove supports and clean up / test-fit the parts. (4)

5.2 Any TWO ✓✓: infill percentage; number of walls / perimeters; layer height; print orientation; material choice. (2)

5.3 Any FOUR tests ✓✓✓✓ with expected results, e.g. sensors read black and white correctly (Serial Monitor shows different values); motors turn the right way for forward / left / right; robot follows a straight line; follows a sharp bend; stops at the finish bar; battery lasts the full race. (4)

5.4 Any TWO ✓✓: no loose wires or short circuits; battery connected with the correct polarity; wheels off the ground / robot on a stand for the first test; all screws tight; switch in the OFF position before connecting the battery. (2)

5.5 Any TWO ✓✓: use more sensors (3 or 5) for smoother following; use proportional control instead of simple on/off turning; lighter chassis; better motors / wheels with more grip; fine-tune the speed on straights and bends. (2)

5.6 $v = s \div t \checkmark = 3 \div 12 = 0{,}25\ m/s \checkmark$ (2)

[16]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE